

Lab 6

```
library IEEE;

use IEEE.STD_LOGIC_1164.ALL;

entity cache_controller is
    Port(
        clk : in STD_LOGIC;
        hit: in STD_LOGIC;
        rread: out STD_LOGIC
    );

end cache_controller;

architecture behavioral of cache_controller is
begin

    process(clk, hit)
        begin
            if rising_edge(clk) then
                if (hit = '1') then
                    rread <= '1';
                else
                    rread <= '0';
                end if;
            end if;
        end process;

    end behavioral;
```

Notes:

- To test, replace entity name from cache_controller into top (due to typical standard of using top instead for main entity)
- Read was replaced with rread as read is a codeword for VHDL

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```
module cache_decoder(  
    input clk,  
    Input [31:0] address,  
    output [19:0] tag,  
    output [7:0] index,  
    Output [3:0] blk_offset  
);  
  
always@(posedge clk)  
begin  
    assign tag = address[31:12];  
    assign index = address[11:4];  
    assign Blk_offset = address[3:0];  
end  
  
endmodule
```